

Gradient of a Scalar Function

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5. Scalar Functions

5.1 In $\mathbb{R}^3$ with coordinates (x, y, z) a function

$f(x, y, z)$
defined for every location (perhaps in a domain) with real numbers as outputs is called a scalar function.

5.2 Examples:

(a) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ is a scalar function

(b) $f(x, y, z) = e^x \sqrt{y} \ln(z)$ is a scalar function
whose domain is

$$\{(x, y, z): x \in \mathbb{R}, y \geq 0, z > 0\}$$

(c) $z = xy$ is not a scalar function, this is an equation.

We can rewrite $z = xy$ as $z - xy = 0$, and say that $f(x, y, z) = z - xy$ is a scalar function, and $f(x, y, z) = 0$ is the same equation as above. Later, we will see that this interpretation adds meaning to the equation

(d) $\vec{f}(x, y, z) = \begin{bmatrix} y \\ z \\ x \end{bmatrix} = y\hat{i} + z\hat{j} + x\hat{k}$ is not a scalar function.

This is a vector field.

(e) $f(x, y) = \frac{x^2 y}{y - x}$ is a scalar function on $\mathbb{R}^2$ with (x, y) coordinates, and domain $\{(x, y): y - x \neq 0\}$

6. Gradient

6.1

Gradient

Gradient

The setting is that we are given a scalar function $f(x, y, z)$ that is defined and differentiable in a domain in 3-space with Cartesian coordinates x, y, z . We denote the **gradient** of that function by $\text{grad } f$ or ∇f (read **nabla** f). Then the gradient of $f(x, y, z)$ is defined as the vector function

$$(1) \quad \text{grad } f = \nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}.$$

6.2 The gradient (nabla) operator:

$$\vec{\nabla} = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

6.3 Example: Find $\vec{\nabla} f$ for $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$

Soln: As only (x, y) is involved, assume $\vec{\nabla} = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix}$.

$$\begin{aligned} \text{Compute: } \frac{\partial}{\partial x} \left(\frac{x^2 - y^2}{x^2 + y^2} \right) &= \frac{(x^2 + y^2) \frac{\partial}{\partial x} (x^2 - y^2) - (x^2 - y^2) \frac{\partial}{\partial x} (x^2 + y^2)}{(x^2 + y^2)^2} \quad (\text{Quotient Rule}) \\ &= \frac{(x^2 + y^2)(2x) - (x^2 - y^2)(2x)}{(x^2 + y^2)^2} = \frac{(2x)(x^2 + y^2 - x^2 + y^2)}{(x^2 + y^2)^2} \\ &= \frac{2x(2y^2)}{(x^2 + y^2)^2} = \frac{4xy^2}{x^4 + 2x^2y^2 + y^4} \end{aligned}$$

$$\frac{\partial}{\partial y} \left(\frac{x^2 - y^2}{x^2 + y^2} \right) = \frac{-4xy}{x^4 + 2x^2y^2 + y^4}$$

$$\text{So } \vec{\nabla} f = \begin{bmatrix} \frac{4xy^2}{x^4 + 2x^2y^2 + y^4} \\ \frac{-4xy}{x^4 + 2x^2y^2 + y^4} \end{bmatrix} = \frac{4xy}{(x^2 + y^2)^2} \begin{bmatrix} y \\ -x \end{bmatrix}$$

$$\left| \frac{\partial f}{\partial x^4 + 2x^2 y^2 + y^4} \right|$$

64 Slide:

Find the gradient of the following functions:

$$f = 9x^2 + 4y^2$$

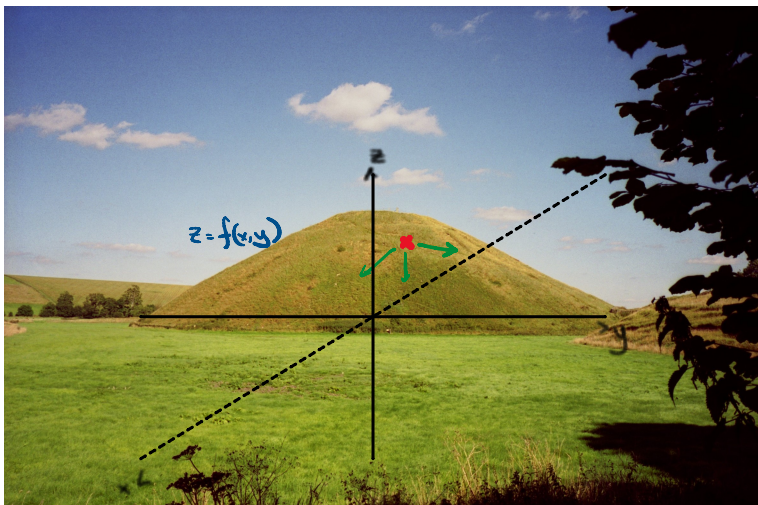
$$f = y/x$$

7. Directional Derivative

7.1 For a unit vector $\vec{u}$, Directional Derivative of a scalar field f in the direction of $\vec{u}$ is given by:

$$D_{\vec{u}} f = \vec{u} \cdot \vec{\nabla} f$$

7.2 Meaning



How much will the height (i.e. $f(x, y)$) change as we run down in different directions?

7.3

ATTENTION! If the direction is given by a vector $\mathbf{a}$ of any length ($\neq 0$), then

(5*)

$$D_{\mathbf{a}} f = \frac{df}{ds} = \frac{1}{|\mathbf{a}|} \mathbf{a} \cdot \text{grad } f.$$

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7.4 Example: Find the derivative in the direction $\vec{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ for

$$f(x, y) = \frac{x}{(x^2 + y^2)} \quad \text{at } P(1, 1)$$

Soln: Compute $\vec{\nabla} f$

$$\frac{\partial}{\partial x} \left(\frac{x}{(x^2 + y^2)} \right) = \frac{(x^2 + y^2) \left(\frac{\partial}{\partial x} x \right) - x \left(\frac{\partial}{\partial x} (x^2 + y^2) \right)}{(x^2 + y^2)^2} = \frac{(x^2 + y^2) - x(2x)}{(x^2 + y^2)^2}$$
$$= \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\frac{\partial}{\partial y} \left(\frac{x}{(x^2 + y^2)} \right) = -\frac{x}{(x^2 + y^2)^2} (2y) = \frac{-2xy}{(x^2 + y^2)^2}$$

$$\text{So } \vec{\nabla} f = \frac{1}{(x^2 + y^2)^2} \begin{bmatrix} y^2 - x^2 \\ -2xy \end{bmatrix}$$

$$\text{at } P(1, 1), \text{ get } \vec{\nabla} f|_{(1,1)} = \vec{\nabla} f \Big|_{(1,1)} = \frac{1}{(1^2 + 1^2)^2} \begin{bmatrix} 1^2 - 1^2 \\ -2(1)(1) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

Now for $\vec{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $|\vec{a}| = \sqrt{2} \neq 1$ (not unit vector)

So directional derivative:

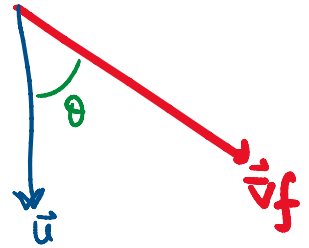
$$\begin{aligned} D_{\vec{a}} f &= \frac{1}{|\vec{a}|} \vec{a} \cdot \vec{\nabla} f \Big|_{(1,1)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{4} \begin{bmatrix} 0 \\ -2 \end{bmatrix} \\ &= \frac{1}{4\sqrt{2}} \left((1)(0) + (1)(-2) \right) \\ &= \frac{-2}{4\sqrt{2}} = \frac{-1}{2\sqrt{2}} \end{aligned}$$

$$= \frac{-2}{4\sqrt{2}} = \frac{-1}{2\sqrt{2}}$$

7.5 From definition of dot-product: (using $|\vec{u}|=1$)

$$D_{\vec{u}}f = \vec{u} \cdot \vec{\nabla}f = |\vec{u}| |\vec{\nabla}f| \cos \theta = |\vec{\nabla}f| \cos \theta$$

We know $-1 < \cos \theta < 1$, and reaches the maximum 1 when $\theta = 0$, so $D_{\vec{u}}f$ reaches a maximum of $|\vec{\nabla}f|$ when $\vec{\nabla}f$ and $\vec{u}$ are in the same direction.



7.6

THEOREM

Use of Gradient: Direction of Maximum Increase

Let $f(P) = f(x, y, z)$ be a scalar function having continuous first partial derivatives in some domain B in space. Then $\text{grad } f$ exists in B and is a vector, that is, its length and direction are independent of the particular choice of Cartesian coordinates. If $\text{grad } f(P) \neq \mathbf{0}$ at some point P , it has the direction of maximum increase of f at P .

Theorem 1

Let $f(x, y, z)$ be a scalar function with continuous first partials in the neighbourhood of some point P . Then:

- (1) $\vec{\nabla}f$ exists and is a vector at P
- (2) length and direction of $\vec{\nabla}f$ is independent of coordinates
- (3) if $\vec{\nabla}f|_P \neq \vec{0}$, then it has the direction of maximum increase of f at P .

7.7 This is a popular idea and is often used to minimize Cost (Economics) or Loss (Machine Learning) functions.

8 - Level Surfaces and the gradient

8.1 An equation of the form $f(x, y, z) = 0$ is called a level surface.
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e.g. The upper-half cone

$$z = \sqrt{x^2 + y^2}$$

can also be written as

$$z^2 = x^2 + y^2 \quad (z \geq 0)$$

or as

$$x^2 + y^2 - z^2 = 0 \quad (z \geq 0)$$

and thus as

$$f(x, y, z) = 0 \quad \text{where} \quad f(x, y, z) := x^2 + y^2 - z^2 \\ \text{for domain } \{(x, y, z) : z \geq 0\}$$

e.g. The circle $x^2 + y^2 = 4$ can be written as

$$g(x, y) = 4 \quad \text{where} \quad g(x, y) := x^2 + y^2$$

$$\text{or as } f(x, y) = 0 \quad \text{where} \quad f(x, y) = x^2 + y^2 - 4$$

In this case the circle is a level curve of both f and g .

In general, $f(x, y, z) = k$ defines a level surface.

8.2 Let's use $f(x, y) = 4$ where $f(x, y) := x^2 + y^2$

$$\text{Then } \vec{\nabla} f = [2x, 2y]$$

Note that we know how to parametrize a circle of radius 2.

$$\vec{r}(t) = [2\cos t, 2\sin t], \quad \text{with } x = 2\cos t, \quad y = 2\sin t$$

$$\text{So tangent vector: } \vec{r}'(t) = [-2\sin t, 2\cos t]$$

$$\text{in other words: } \vec{v} = [-y, x]$$

We find that

$$\vec{v} \cdot \vec{\nabla} f = \begin{bmatrix} -y \\ x \end{bmatrix} \cdot \begin{bmatrix} 2x \\ 2y \end{bmatrix} = (-y)(2x) + (x)(2y) = 0$$

for all x, y .

In other words, $\vec{\nabla} f$ is normal to the curve $f(x, y) = 4$.

8.3

THEOREM 2

Gradient as Surface Normal Vector

Let f be a differentiable scalar function in space. Let $f(x, y, z) = c = \text{const}$ represent a surface S . Then if the gradient of f at a point P of S is not the zero vector, it is a normal vector of S at P .

8.4

EXAMPLE 2

Gradient as Surface Normal Vector. Cone

Find a unit normal vector $\mathbf{n}$ of the cone of revolution $z^2 = 4(x^2 + y^2)$ at the point $P: (1, 0, 2)$.

Solution. The cone is the level surface $f = 0$ of $f(x, y, z) = 4(x^2 + y^2) - z^2$. Thus (Fig. 217)

$$\text{grad } f = [8x, 8y, -2z], \quad \text{grad } f(P) = [8, 0, -4]$$

$$\mathbf{n} = \frac{1}{|\text{grad } f(P)|} \text{grad } f(P) = \left[\frac{2}{\sqrt{5}}, 0, -\frac{1}{\sqrt{5}} \right].$$

$\mathbf{n}$ points downward since it has a negative z -component. The other unit normal vector of the cone at P is $-\mathbf{n}$. ■

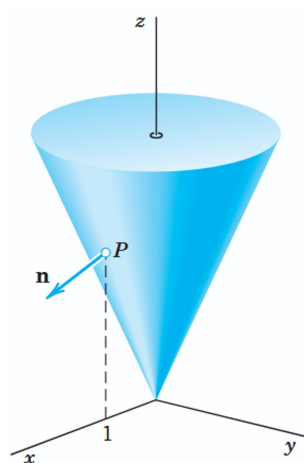


Fig. 217. Cone and unit normal vector $\mathbf{n}$

Note: It's not at all obvious that $\vec{\nabla} f$ is normal to $f(x, y, z) = c$ when we are dealing with 2-D surfaces in $\mathbb{R}^3$. We need to learn how to parametrize surfaces.